

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 3: Landing Gear, Wheels, Tyres & Brakes

Syllabus Source: CAE Oxford (pp. 99-148) | Pass Target: >= 90% AviationExam

1. OLEO-PNEUMATIC SHOCK STRUT MECHANICS

Modern transport aircraft undercarriages use oleo-pneumatic shock absorber struts. The strut absorbs the kinetic energy of landing touchdown and dissipates it, while providing resilient cushioning during surface taxiing:

- Operating Mediums: Strut contains two distinct fluids: 1. Hydraulic fluid (oil) in the lower cylinder; 2. Compressed dry nitrogen gas in the upper chamber.
- Compression Stroke (Touchdown): During landing impact, the strut telescopes. The lower piston forces hydraulic oil upward through a small calibrated metering orifice into the upper cylinder, compressing the nitrogen. Oil forced through the restricted orifice transforms kinetic energy into heat (dissipated damping), while nitrogen compression cushions the impact.
- Rebound Damping: A recoil valve or tapered metering pin restricts the return flow of oil as the strut extends, preventing violent spring-back oscillations.
- Maintenance Servicing Order: CRITICAL EXAM POINT: Oleo struts must always be serviced with HYDRAULIC OIL FIRST until full to the level plug, and then inflated with DRY NITROGEN to achieve the specified chrome extension height!

2. RETRACTION SYSTEMS, LOCKS & EMERGENCY EXTENSION

Landing gears are retracted hydraulically into aerodynamically sealed wheel wells in flight, locked mechanically, and provided with emergency extension backup:

- Uplocks & Downlocks: Hydraulic actuators hold the gear in transit. Once fully retracted or extended, over-center mechanical locks (uplocks and downlocks) engage. The gear is held down by MECHANICAL LOCKS, not hydraulic pressure!
- Ground Safety Pins: Mechanical lock pins inserted into gear drag struts while parked on ground to physically prevent accidental gear retraction.
- Emergency Extension Methods: 1. Gravity Free-Fall: Uplocks are mechanically or electrically released; gear doors open and the gear falls and locks down under gravity and aerodynamic airflow; 2. Emergency pneumatic nitrogen blow-down.
- Nose Wheel Centering: Nose gear shock struts feature an internal mechanical centering cam. As the nose strut extends after take-off, the internal cams align the nose wheels perfectly fore-and-aft before entering the wheel well, preventing catastrophic jamming.

3. AIRCRAFT WHEELS, TYRES & THERMAL FUSIBLE PLUGS

Feature	Engineering Design	Operational & Safety Rule
Wheel Construction	Two-piece split-hub design made of forged aluminum or magnesium alloy.	Sealed by an elastomer O-ring. Assembled with tie-bolts. Always deflate tyre before loosening wheel tie-bolts!

Thermal Fusible Plugs	Hollow threaded brass bolts filled with a eutectic metal alloy plug.	Melt at a predetermined high temperature (~150°C to 180°C) resulting from heavy or rejected take-off (RTO) braking. Releases tyre nitrogen pressure slowly into the brake assembly, PREVENTING CATASTROPHIC TYRE EXPLOSION and shrapnel damage!
Tyre Inflation Gas	Nitrogen is mandatory for all commercial transport aircraft tyres.	Nitrogen is inert and contains no moisture; will not support combustion, maintains stable pressure across temperature extremes, and prevents oxidation of the inner tyre carcass rubber.
Tyre Wear & Creep	Chine tyres used on nose wheels to deflect runway water away from rear-mounted jet engines.	Tyre creep (slipping of tyre around the rim under heavy braking) is monitored by matching white creep marks painted across tyre and rim.

4. HYDROPLANING (AQUAPLANING) FORMULAS & PHENOMENA

Hydroplaning occurs when aircraft tyres become separated from the paved runway surface by a continuous fluid film of water, reducing braking coefficient of friction and directional control to virtually zero:

Hydroplaning Type	Physics & Water Depth	Critical Speed Formula (Knots)
Dynamic Hydroplaning	Standing water on runway (> 3 mm depth). Water inertia builds up a hydrodynamic wedge that lifts tyre completely off pavement.	Rotating Tyre: $V_p = 9 \times \sqrt{P}$ [psi] Non-Rotating (Skidding): $V_p = 7.7 \times \sqrt{P}$ [psi]
Viscous Hydroplaning	Very thin film of moisture (< 0.1 mm) on smooth pavement (paint markings, rubber deposits). Oil and dust create a soapy lubricating film.	Can occur at MUCH LOWER SPEEDS than dynamic hydroplaning; persists down to low taxi speeds!
Reverted Rubber	Prolonged wheel lockup during skid. Friction heat boils trapped water into high-pressure steam, vulcanizing and burning tyre rubber into soft white seal.	Steam cushion supports aircraft; tyre shows distinctive burnt/scalded oval rubber patch.

[!] EXAM TRAP (AVIATIONEXAM) - Dynamic Hydroplaning Calculation Examples

- Example 1 (Rotating Tyre): Tyre pressure = 144 psi. Dynamic hydroplaning speed = $9 \times \sqrt{144} = 9 \times 12 = 108$ KNOTS!
- Example 2 (Non-Rotating Tyre / Spin-up on Touchdown): Tyre pressure = 100 psi. Speed = $7.7 \times \sqrt{100} = 7.7 \times 10 = 77$ KNOTS!
- Note: A skidding (locked) wheel hydroplanes at a LOWER speed (7.7) than a rolling wheel (9.0)!

5. ANTI-SKID SYSTEMS & CARBON BRAKE PACKS

Modern multi-disc brakes use multiple rotating discs keyed to the wheel and stationary stator discs keyed to the axle torque tube. Carbon brake discs provide superior heat capacity, lighter weight, and wear less at high operating temperatures compared to steel brakes:

- Anti-Skid Operating Principle: Wheel speed transducers monitor rotational deceleration. If wheel deceleration exceeds a predetermined threshold (indicating imminent skid), the anti-skid valve dumps hydraulic brake pressure instantly, allowing the wheel to spin back up, then smoothly reapplies modulated pressure.

- Anti-Skid Functions: 1. Normal skid control; 2. Touchdown protection (prevents brake pressure application before touchdown, even if pilot presses pedals, until wheel spin-up reaches 30-50 kt or squat switch signals ground mode); 3. Locked wheel crossover protection; 4. Gear retraction braking (automatically applies low pressure to stop wheel rotation before entering wheel well).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Hydroplaning Speed Calculations

Formulas: - Dynamic Hydroplaning (Knots) = $9 \times \sqrt{\text{Tire Pressure in PSI}}$. -

Dynamic Hydroplaning (Spin-down) = $7.7 \times \sqrt{\text{Tire Pressure in PSI}}$. Types: 1.

Dynamic: Standing water separates tire from runway. 2. Viscous: Thin film of oil/dust on smooth runway (occurs at much lower speeds). 3. Reverted Rubber: Locked wheel skids, boiling moisture into steam cushions.

6. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the primary purpose of thermal fusible plugs installed in aircraft wheels?

- [A]: To melt and rapidly inflate the tyre during high-speed take-offs
- [B]: To melt and release tyre pressure safely if brake and wheel temperatures exceed limits, preventing catastrophic tyre explosion
- [C]: To lock the brakes when parked
- [D]: To indicate brake pad wear limits

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fusible plugs contain a low-melting-point alloy (150-180°C). During severe braking or an RTO, high brake heat conducts to the wheel rim; the plug melts, venting tyre nitrogen harmlessly before the tyre explodes.

Q2: An aircraft has a main tyre pressure of 144 psi. What is the calculated dynamic hydroplaning speed for a runway with standing water?

- [A]: 77 kt
- [B]: 95 kt
- [C]: 108 kt
- [D]: 144 kt

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula for rotating tyre: $V_p = 9 \times \sqrt{P} = 9 \times \sqrt{144} = 9 \times 12 = 108$ knots.

Q3: In an oleo-pneumatic shock absorber strut, what primary medium absorbs and dissipates the initial impact energy?

- [A]: Nitrogen gas compression
- [B]: Hydraulic oil forced through a restricted metering orifice
- [C]: Steel coil springs
- [D]: Rubber snubber blocks

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Nitrogen compression stores energy and provides spring cushioning, but the initial impact energy is absorbed and DISSIPATED as heat by hydraulic oil being forced through the calibrated metering orifice.

Q4: Why are commercial aircraft tyres inflated with dry nitrogen rather than compressed ambient air?

- [A]: Nitrogen is lighter than air
- [B]: Nitrogen is completely inert, dry (prevents rim corrosion and pressure fluctuations), and eliminates the risk of tyre explosion at high temperatures
- [C]: Nitrogen increases braking friction
- [D]: Nitrogen is cheaper than air

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Nitrogen contains no oxygen or moisture. It prevents rim corrosion, avoids high internal steam pressure rises from moisture at high brake temperatures, and eliminates internal combustion/explosion risks.

Q5: How does the anti-skid 'touchdown protection' feature function?

- [A]: It deploys the ground spoilers automatically
- [B]: It inhibits brake application until the main wheels have spun up to a minimum speed or ground squat switches confirm weight on wheels
- [C]: It lowers the nose wheel gently to the runway
- [D]: It disconnects nose wheel steering

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Touchdown protection prevents hydraulic pressure from reaching the brakes even if the pilot inadvertently presses the brake pedals before touchdown, ensuring the tyres do not blow out upon contacting the runway.